

First Draft, everything is subject to change, just what I've thought of so far

UI Mockups:

```
-----
|           Score: 120      Time: 00:45           |
|-----|
| #####|
| # S . W . R . . . R . . . . W . . W . P . W . . #|
| # . W . . W . P . . W . . R . . W . . . . W . . . . B #|
| # . . . R . . . . W . . . . . W . . W . P . W . . E . #|
| #####|
|-----|
```

Elements:

- S = Start
- E = Exit
- R = Regular Reward
- B = Bonus Reward
- P = Punishment
- W = Wall
- Enemies would be moving
- Player character

At the top:

- Score display
- Timer display

Ui for Start Screen:

2d Arcade Escape

[Start Game]
[Instructions]
[Exit]

UML Class Diagrams:

What's needed?

Classes, Attributes, Methods, Relationships

Game:

Game:

- Board: board
- Player: player
- Enemies: List<enemies>
- Score: int
- Timer: timer

Methods:

- start()
- update()
- checkWin()
- checkLoss()

Board:

Board:

- Grid: Cell()
- Width: int
- Height: int

Methods:

- loadMap()
- isValidMove()
- getCell()

Cell:

Cell:

- Position: position
- Entity: GameObject

GameObject(superclass):

GameObject(abstract):

- Position: position

Methods:

- update()

Player(extension of GameObject):

Player:

- Score: int

Methods:

- move(Direction)
- collectReward()
- applyPenalty()

Enemy (abstract) extension of GameObject:

Enemy:

- move()

MovingEnemy (extension of Enemy):

- chasePlayer()

Punishment (extension of Enemy):

- No movement

Reward:

Reward:

- Value: int
- RegularReward (extension of Reward)
- BonusReward (extension of Reward)

Position:

Position:

- X: int
- Y: int

Ideas on how to draw the UML:

- Can have it drawn out by hand
- Powerpoint
- [Draw.io](https://draw.io)

The diagram of the system structure could look something like

```
-----  
Player  
-----  
- score: int  
- position: Position  
-----  
+ move()  
+ collectReward()  
-----
```